Appendix

This appendix provides supplementary materials for **MambaRGS: Efficient Referring Grasp Synthesis via Text-Image Cross-State Modulation and Rotation-Aware Decoding**. It includes additional dataset details, experimental results, ablation studies, efficiency analysis, and limitations that could not be included in the RA-L manuscript because of the page limit.

1. GraspNet-VLG Dataset Details

GraspNet-VLG is built upon GraspNet-1Billion and adapts its cluttered RGB-D scenes, object categories, instance masks, and grasp annotations to the referring grasp synthesis setting. Each data entry associates a specific RGB-D image, a target object instance, a segmentation mask, natural language referring expressions, and planar grasp annotations. As summarized in Table 1, existing datasets usually focus on either grasp detection or visual grounding alone. For example, GraspNet provides large-scale real-scene grasp annotations but does not contain referring expressions, while RoboRefIt contains abundant language expressions but lacks grasp annotations. Compared with OCID-VLG, which also supports language-guided grasping, GraspNet-VLG provides more object categories and RGB-D images under real cluttered scenes. Therefore, GraspNet-VLG serves as a complementary benchmark for evaluating language-guided robotic grasping in realistic multi-object scenarios.

Table 1: Comparison of representative public datasets for grasp detection, visual grounding, and language-guided robotic grasping.

Dataset	Scene	Grasps	RefExp	Categories	Images
GraspNet	Real	✓	×	88	97K
OCID-Grasp	Real	✓	×	31	1.7K
OV-Grasp	Both	×	63K	117	/
RoboRefIt	Real	×	50.7K	66	10.7K
GraspAnything++	Synthetic	✓	10M	236	994K
OCID-VLG	Real	✓	89.6K	31	1.7K
GraspNet-VLG	Real	✓	106K	87	2.5K

Since GraspNet provides 6D grasp poses rather than directly usable 2D planar grasp rectangles, the 6D grasp annotations are projected into the image plane and converted into 2D grasp rectangles. A systematic quality-control pipeline is then applied to the projected grasp labels. Specifically, we remove grasp rectangles whose centers fall outside the target object mask, remove samples with very small target masks, discard samples with too few valid grasps, repair overly long grasp rectangles according to the target-mask support region, and apply center-distance and angle-aware non-maximum suppression to reduce highly redundant grasp candidates. This refinement is performed without using model predictions. The effect of this annotation

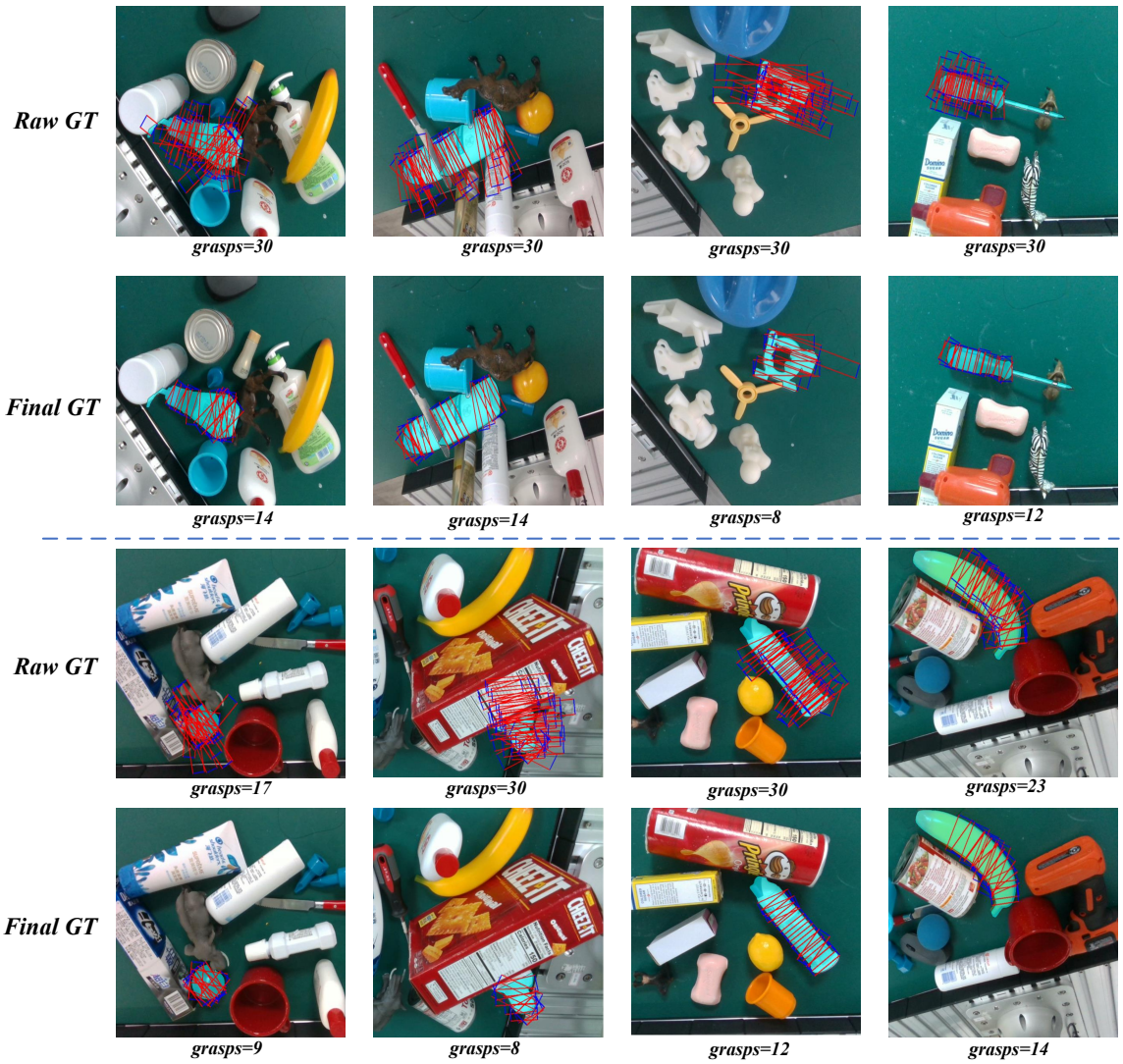


Figure 1: Qualitative comparison of original and refined grasp annotations in GraspNet-VLG. Top row: original grasp ground-truth annotations generated from projected 6D grasp poses. Bottom row: final refined grasp ground-truth annotations.

refinement is illustrated in Fig. 1. The original projected annotations may contain invalid or noisy grasp rectangles, such as grasps located outside the target object region, overly long rectangles that exceed the object support area, and highly redundant candidates with similar centers and orientations. After refinement, the retained grasp annotations are more compact, object-aligned, and visually consistent with the target masks, which provides cleaner supervision for both referring object segmentation and grasp prediction.

After filtering, the corrected GraspNet-VLG benchmark contains **2,500 RGB-D images**, **17,813 image-target-mask samples**, **106K referring expressions**, **87 valid object categories**, and **163,466 valid grasp rectangles**. The only category fully removed is ($object_id$=28), i.e., toy airplane, due to insufficient valid annotations after filtering. These statistics further demonstrate that GraspNet-VLG retains the real-scene diversity of GraspNet-1Billion while introducing language-conditioned target specification and refined planar

grasp supervision for referring grasp synthesis.

2. Additional Experimental Results

2.1. TICM Operator Ablation

To isolate the contribution of the Mamba/SS2D design from the three-branch formulation itself, we keep the TICM inputs unchanged and replace the internal branch modeling operator. The compared variants use the same visual context, sentence-level global semantics, and token-level text-image correspondence branches. As shown in Table 2, the SS2D-based TICM achieves the best performance across all RIS and RGS metrics while using fewer parameters than the CNN- and self-attention-based alternatives. Compared with the three-branch CNN variant, the SS2D-based TICM improves mIoU from 93.44 to 95.23, J@1 from 87.90 to 90.72, and J@Any from 91.52 to 94.04, with gains of 1.79, 2.82, and 2.52 points, respectively. Compared with the three-branch self-attention variant, it improves mIoU by 1.48 points, J@1 by 2.65 points, and J@Any by 2.58 points. In addition, the SS2D-based variant uses 218.36M parameters, which is 5.24M fewer than the CNN variant and 19.18M fewer than the self-attention variant. These results suggest that the gain of TICM comes not only from the three complementary branches, but also from SS2D-based state modeling within those branches.

Table 2: Controlled ablation on the branch modeling operator in TICM. All variants use the same three-branch inputs.

Variant	RIS						RGS		Param. (M)
	mIoU	Pr@50	Pr@60	Pr@70	Pr@80	Pr@90	J@1	J@Any	
Three-branch CNN	93.44	96.10	95.85	95.63	95.30	94.29	87.90	91.52	223.60
Three-branch self-attention	93.75	96.28	96.13	96.06	95.90	94.95	88.07	91.46	237.54
Three-branch SS2D	**95.23**	**98.20**	**97.64**	**97.29**	**96.85**	**95.49**	**90.72**	**94.04**	**218.36**

2.2. Internal Ablation of TICM

We further analyze the internal design of TICM, including branch contribution, sequential modulation, and learnable gates. As shown in Table 3, all reduced variants are inferior to the full model, confirming the effectiveness of the complete TICM design. For branch contribution, the Visual + Global variant obtains 93.30 mIoU, 88.65 J@1, and 91.59 J@Any, while the Visual + Correlation variant achieves 94.07 mIoU, 90.48 J@1, and 93.19 J@Any. Compared with Visual + Global, Visual + Correlation improves mIoU by 0.77 points, J@1 by 1.83 points, and J@Any by 1.60 points, indicating that token-level text-image correlation is particularly helpful for fine-grained language-conditioned grounding and grasp prediction. Nevertheless, the full model further improves over Visual + Correlation by 1.16 mIoU points, 0.24 J@1 points, and 0.85 J@Any

points, showing that the global semantic branch and correlation branch provide complementary information. For modulation strategy, removing the feedback from the updated visual state to the language-derived states leads to the Unidirectional Modulation variant, which obtains 94.86 mIoU, 90.26 J@1, and 93.58 J@Any. The full model improves these results to 95.23 mIoU, 90.72 J@1, and 94.04 J@Any, with gains of 0.37, 0.46, and 0.46 points, respectively. This verifies the benefit of sequential cross-state modulation. For gate design, the Fixed Gates variant obtains 94.18 mIoU, 90.53 J@1, and 93.65 J@Any. By learning the gate weights adaptively, the full model improves mIoU by 1.05 points, J@1 by 0.19 points, and J@Any by 0.39 points. These results indicate that learnable gates help TICM adaptively balance visual, global-semantic, and local-correlation interactions.

Table 3: Internal ablation study of TICM on the OCID-VLG dataset.

Variant	**RIS**						**RGS**	
	mIoU	**Pr@50**	**Pr@60**	**Pr@70**	**Pr@80**	**Pr@90**	**J@1**	**J@Any**
Visual + Global	93.30	95.79	95.45	95.35	95.08	93.88	88.65	91.59
Visual + Correlation	94.07	97.61	97.42	96.31	96.01	94.84	90.48	93.19
Unidirectional Modulation	94.86	97.94	97.79	96.55	96.79	95.14	90.26	93.58
Fixed Gates	94.18	97.55	97.49	96.19	95.84	94.66	90.53	93.65
Full Model	**95.23**	**98.20**	**97.64**	**97.29**	**96.85**	**95.49**	**90.72**	**94.04**

2.3. ARC Kernel Sensitivity and Angular Robustness

The final grasp angle is predicted by the dense grasp head and supervised through the standard $\sin(2\theta)$ and $\cos(2\theta)$ maps. The rotation angle predicted in ARC is a feature-level orientation-adaptation parameter. Its role is to rotate the convolutional receptive pattern so that the decoder can better respond to oriented local structures, such as object boundaries, elongated parts, and graspable regions.

As shown in Table 4, replacing the standard convolutional decoder with ARC consistently improves the performance. Compared with the Std. Conv variant, ARC with $N = 1$ improves mIoU from 93.27 to 95.23, J@1 from 89.14 to 90.72, and J@Any from 92.17 to 94.04, corresponding to gains of 1.96, 1.58, and 1.87 points, respectively. This confirms that adaptive kernel rotation is beneficial for orientation-sensitive dense decoding in referring grasp synthesis. Increasing the number of ARC kernels further improves several segmentation metrics. For example, $N = 2$ improves mIoU to 95.66 and Pr@90 to 96.14, while $N = 4$ achieves the best mIoU of 95.78, Pr@50 of 98.42, Pr@60 of 98.26, Pr@80 of 97.66, and J@Any of 94.12. However, larger N does not improve the top-ranked grasp accuracy. The best J@1 is obtained when $N = 1$, reaching 90.72, whereas $N = 2$ and $N = 4$ obtain 89.48 and 89.10, respectively. Since robotic execution usually selects the top-ranked grasp candidate, J@1 is particularly important for real deployment. Therefore, we adopt $N = 1$

in the final model, which provides the best top-ranked grasp performance while still substantially improving segmentation and overall grasp feasibility over standard convolution.

Table 4: Sensitivity analysis of the number of ARC kernels on the OCID-VLG dataset.

RAD Variant	mIoU	Pr@50	Pr@60	Pr@70	Pr@80	Pr@90	J@1	J@Any
Std. Conv	93.27	96.10	95.76	95.59	94.97	93.11	89.14	92.17
N=1	95.23	98.20	97.64	97.29	96.85	95.49	**90.72**	94.04
N=2	95.66	98.33	98.19	**98.06**	97.53	**96.14**	89.48	93.27
N=4	**95.78**	**98.42**	**98.26**	98.00	**97.66**	96.05	89.10	**94.12**

We further evaluate whether the proposed orientation-sensitive decoding improves robustness under stricter angular constraints. As shown in Table 5, when the angular threshold is tightened from 30° to 5°, our method shows a smaller performance drop than ETRG on both OCID-VLG and GraspNet-VLG. On OCID-VLG, ETRG drops from 89.06 J@1 at 30° to 74.51 J@1 at 5°, corresponding to a relative drop of 16.34% and a retention of 83.66%. In contrast, our method drops from 90.72 to 84.12, with a relative drop of only 7.28% and a retention of 92.72%. This means that our method retains 9.06 percentage points more of its original J@1 performance than ETRG under the strictest angular threshold. On GraspNet-VLG, our method also achieves better angular robustness, retaining 54.19% of its original J@1 performance, compared with 51.20% for ETRG. These results indicate that ARC-based rotation-aware decoding helps produce more stable top-ranked grasp orientations, especially when the evaluation requires high angular precision.

Table 5: Angular robustness analysis of J@1 when tightening λ_θ from 30° to 5°, where Relative Drop = $(J_{30^\circ} - J_{5^\circ})/J_{30^\circ} \times 100\%$ and Retention = $J_{5^\circ}/J_{30^\circ} \times 100\%$.

Dataset	Method	J@1 at 30°	J@1 at 5°	Relative Drop ↓	Retention ↑
OCID-VLG	ETRG	89.06	74.51	16.34%	83.66%
OCID-VLG	**Ours**	**90.72**	**84.12**	**7.28%**	**92.72%**
GraspNet-VLG	ETRG	86.49	44.28	48.80%	51.20%
GraspNet-VLG	**Ours**	**88.54**	**47.98**	**45.81%**	**54.19%**

2.4. Robustness Analysis on GraspNet-VLG

The main manuscript reports robustness analysis on OCID-VLG. Here, we provide the corresponding compact robustness analysis on the GraspNet-VLG benchmark. We evaluate robustness from two aspects: spatial overlap robustness by varying λ_{IoU}, and angular robustness by varying λ_θ. The compared methods include CROG, ETRG, and MambaRGS.

Table 6 reports the results when the IoU threshold is varied from 0.25 to 0.5 while the angular threshold is fixed at $\lambda_\theta = 30°$. As the IoU threshold becomes stricter, all methods show performance degradation. However, MambaRGS consistently achieves the best J@1 and J@Any under all IoU thresholds. At the standard threshold $\lambda_{\text{IoU}} = 0.25$, MambaRGS achieves 88.54 J@1 and 96.22 J@Any, outperforming ETRG by 2.05 and 1.21 points, respectively. When the threshold is increased to 0.4, the gains become larger: MambaRGS obtains 76.29 J@1 and 91.12 J@Any, exceeding ETRG by 3.90 and 2.92 points. Under the strictest threshold $\lambda_{\text{IoU}} = 0.5$, MambaRGS still achieves 60.37 J@1 and 82.29 J@Any, outperforming ETRG by 6.33 and 5.92 points, and substantially surpassing CROG by 34.02 and 36.16 points. These results indicate that MambaRGS produces grasp rectangles with better spatial consistency under stricter overlap requirements.

Table 6: Robustness analysis on GraspNet-VLG by varying λ_{IoU} for the Jacquard metric. The angular threshold is fixed at $\lambda_\theta = 30°$.

Method	J@1				J@Any			
	0.25	0.3	0.4	0.5	0.25	0.3	0.4	0.5
CROG [1]	84.64	78.00	52.97	26.35	94.51	90.81	73.15	46.13
ETRG [2]	86.49	82.93	72.39	54.04	95.01	93.69	88.20	76.37
Ours	**88.54**	**85.45**	**76.29**	**60.37**	**96.22**	**94.65**	**91.12**	**82.29**

Table 7 reports the results when the angular threshold is tightened from 30° to 5° while the IoU threshold is fixed at $\lambda_{\text{IoU}} = 0.25$. MambaRGS again achieves the best performance under all angular thresholds. At $\lambda_\theta = 30°$, it obtains 88.54 J@1 and 96.22 J@Any, outperforming ETRG by 2.05 and 1.21 points. When the angular threshold is tightened to 15°, MambaRGS achieves 79.85 J@1 and 91.87 J@Any, exceeding ETRG by 4.23 and 2.86 points. Under the strictest angular threshold of 5°, MambaRGS reaches 47.98 J@1 and 73.46 J@Any, outperforming ETRG by 3.70 and 6.78 points, and CROG by 6.25 and 5.07 points. These results show that the proposed rotation-aware decoding improves the stability of grasp orientation estimation under strict angular constraints.

Table 7: Robustness analysis on GraspNet-VLG by varying λ_θ for the Jacquard metric. The IoU threshold is fixed at $\lambda_{\text{IoU}} = 0.25$.

Method	J@1				J@Any			
	30°	25°	15°	5°	30°	25°	15°	5°
CROG [1]	84.64	82.60	74.80	41.73	94.51	93.39	88.99	68.39
ETRG [2]	86.49	84.42	75.62	44.28	95.01	93.72	89.01	66.68
Ours	**88.54**	**87.11**	**79.85**	**47.98**	**96.22**	**95.21**	**91.87**	**73.46**

Overall, the additional GraspNet-VLG robustness analysis confirms that MambaRGS remains more stable than the compared baselines under both stricter spatial overlap and stricter angular constraints. The increasing

Table 8: Comparison with multi-stage baselines on RIS and RGS metrics. * indicates multi-stage baselines that use GR-ConvNet pretrained on the Jacquard dataset as the grasp synthesis network.

Method	RIS						RGS	
	mIoU	Pr@50	Pr@60	Pr@70	Pr@80	Pr@90	J@1	J@Any
GT-Grounding*	/	/	/	/	/	/	28.7	70.2
GT-Masks+CLIP*	35.00	35.00	35.00	35.00	35.00	35.00	11.9	26.8
SAM+CLIP*	25.70	29.30	28.50	27.40	22.70	9.10	7.2	12.7
GLIP+SAM*	30.30	34.70	34.10	33.50	28.60	11.70	10.7	21.8
Det-Seg+CLIP	29.00	27.20	20.90	17.50	17.20	16.00	28.1	39.2
SSG+CLIP	33.60	35.60	35.60	35.50	35.50	32.80	33.5	34.7
Ours	**95.23**	**98.20**	**97.64**	**97.29**	**96.85**	**95.49**	**90.72**	**94.04**

performance gap under high IoU thresholds suggests better localization consistency of the predicted grasp rectangles, while the advantage under tight angular thresholds supports the effectiveness of orientation-sensitive decoding for grasp angle prediction.

2.5. Multi-Stage Baseline Comparison

We compare MambaRGS with multi-stage baselines that first perform language-guided object grounding and then use a grasp synthesis network for mask-conditioned grasp prediction. As shown in Table 8, MambaRGS substantially outperforms all multi-stage baselines on both RIS and RGS metrics.

The Jacquard-pretrained GR-ConvNet baselines show limited transferability to OCID-VLG. Even with ground-truth grounding, GT-Grounding* only achieves 28.7 J@1 and 70.2 J@Any, indicating that a pretrained grasp detector alone does not transfer well to the referring grasp synthesis setting. When predicted masks are used, the performance drops further: SAM+CLIP* obtains only 7.2 J@1 and 12.7 J@Any, while GLIP+SAM* obtains 10.7 J@1 and 21.8 J@Any. The supervised Det-Seg+CLIP and SSG+CLIP baselines improve the grasping results to 28.1/39.2 and 33.5/34.7 in J@1/J@Any, respectively, but remain far below MambaRGS. In comparison, our method achieves 95.23 mIoU, 90.72 J@1, and 94.04 J@Any, demonstrating the advantage of jointly optimizing object grounding and grasp prediction in an end-to-end RGS framework.

2.6. Resolution-Scaling Efficiency Analysis

The main manuscript reports architectural and efficiency comparison at 416×416 resolution. Here, we further compare the forward FLOPs of MambaRGS and the Transformer-based fusion baseline LAVT-RIS under multiple input resolutions. As shown in Table 9 and Fig. 2, MambaRGS consistently requires fewer

Table 9: Resolution-scaling comparison of forward FLOPs between MambaRGS and LAVT-RIS.

Resolution	Ours FLOPs (G)	LAVT-RIS FLOPs (G)
416×416	375.13	445.55
480×480	499.77	568.02
640×640	885.15	1000.98
1024×1024	2262.86	2521.11
2048×2048	9045.45	9925.96

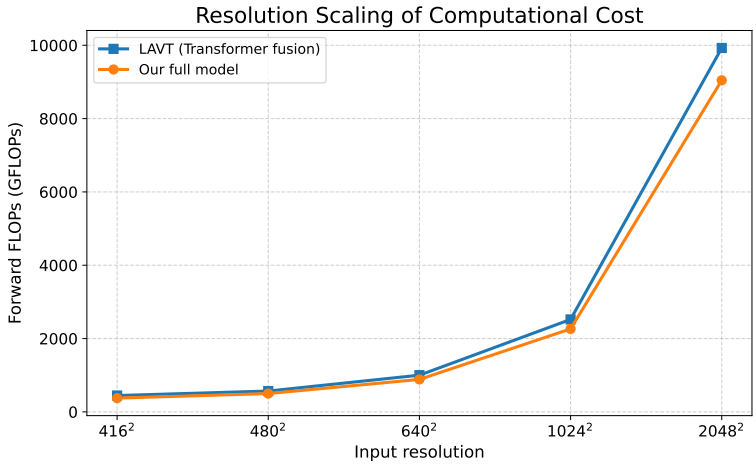


Figure 2: Resolution-scaling comparison of forward FLOPs between our model and the Transformer-based fusion baseline LAVT-RIS. Both methods exhibit approximately linear scaling with respect to the number of input pixels, while our model consistently requires fewer FLOPs across all tested resolutions.

FLOPs than LAVT-RIS across all tested resolutions. At the standard input resolution of 416×416, MambaRGS requires 375.13G FLOPs, while LAVT-RIS requires 445.55G FLOPs. When the resolution increases to 2048×2048, the FLOPs of MambaRGS and LAVT-RIS increase to 9045.45G and 9925.96G, respectively. Both methods show approximately linear scaling with respect to the number of input pixels, but MambaRGS maintains a lower computational cost throughout the full resolution range. This supports the efficiency advantage of the proposed design over the Transformer-based fusion baseline.

Table 10: Real-world wrist-camera evaluation on visually similar objects. Object grounding is evaluated by mIoU, and grasp detection is evaluated by J@1.

Methods	Grounding mIoU (%)	Grasp Detection J@1 (%)
ETRG	76.53	79.20
Ours	**81.64**	**82.67**



Scene1 *Scene2* *Scene3* *Scene4*

Figure 3: Representative real-world scenes with duplicated or visually similar objects for fine-grained referring grasping evaluation.

2.7. Real-World Wrist-Camera Evaluation on Visually Similar Objects

To evaluate fine-grained referring grasping under visually ambiguous scenes, we collected 10 real tabletop scenes using the robot wrist-mounted camera. These scenes contain visually similar or repeated objects, such as duplicated bananas, toothpaste boxes, and soap boxes. The language expressions involve object categories and spatial relations, requiring the model to distinguish the referred object from visually similar instances. Representative examples are shown in Fig. 3. Because the Franka robot platform was temporarily unavailable during revision, this evaluation reports object grounding and grasp detection metrics rather than additional physical execution trials. As reported in Table 10, MambaRGS achieves 81.64 mIoU and 82.67 J@1, outperforming ETRG by 5.11 points in grounding mIoU and 3.47 points in grasp detection J@1. These results indicate that MambaRGS can better distinguish visually similar target objects under attribute- and spatial-relation-based referring expressions.

We further provide qualitative comparisons in Fig. 4. Duplicated or visually similar objects introduce clear ambiguity, especially when the target is specified by spatial relations rather than category names alone. In challenging cases, ETRG produces erroneous responses on non-target toothpaste boxes, while MambaRGS localizes the referred targets more accurately and generates more consistent grasp predictions on the intended objects.

2.8. Failure Cases and Limitations

Benchmark-level challenges. Although MambaRGS achieves strong performance on OCID-VLG and GraspNet-VLG, realistic referring grasp synthesis remains challenging. Visually similar or duplicated objects can still confuse language-conditioned object grounding, especially when targets differ only by subtle attributes or spatial relations. The wrist-camera evaluation above shows that both ETRG and MambaRGS may produce partial responses to non-target objects in ambiguous scenes. More challenging benchmarks with duplicated objects, subtle attributes, and spatial-relation reasoning would further benefit future research.

Model-level limitations. MambaRGS currently uses RGB images and referring expressions as network

inputs. Depth is only used after network prediction in real-robot execution to back-project the predicted 2D grasp rectangle into a 3D grasp pose. Since depth provides explicit geometric and spatial cues, incorporating RGB-D fusion into MambaRGS is a promising direction. In addition, although MambaRGS requires fewer FLOPs than the Transformer-based LAVT-RIS baseline, it is still slower than efficient CNN-based methods such as CROG and ETRG. Future work may explore lightweight CNN-Mamba hybrid architectures to improve inference speed while maintaining strong grounding and grasp prediction accuracy.

References

[1] Georgios Tziafas, XU Yucheng, Arushi Goel, Mohammadreza Kasaei, Zhibin Li, and Hamidreza Kasaei. Language-guided robot grasping: Clip-based referring grasp synthesis in clutter. In *7th Annual Conference on Robot Learning*, 2023.

[2] Houjian Yu, Mingen Li, Alireza Rezazadeh, Yang Yang, and Changhyun Choi. A parameter-efficient tuning framework for language-guided object grounding and robot grasping. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 14353–14360. IEEE, 2025.

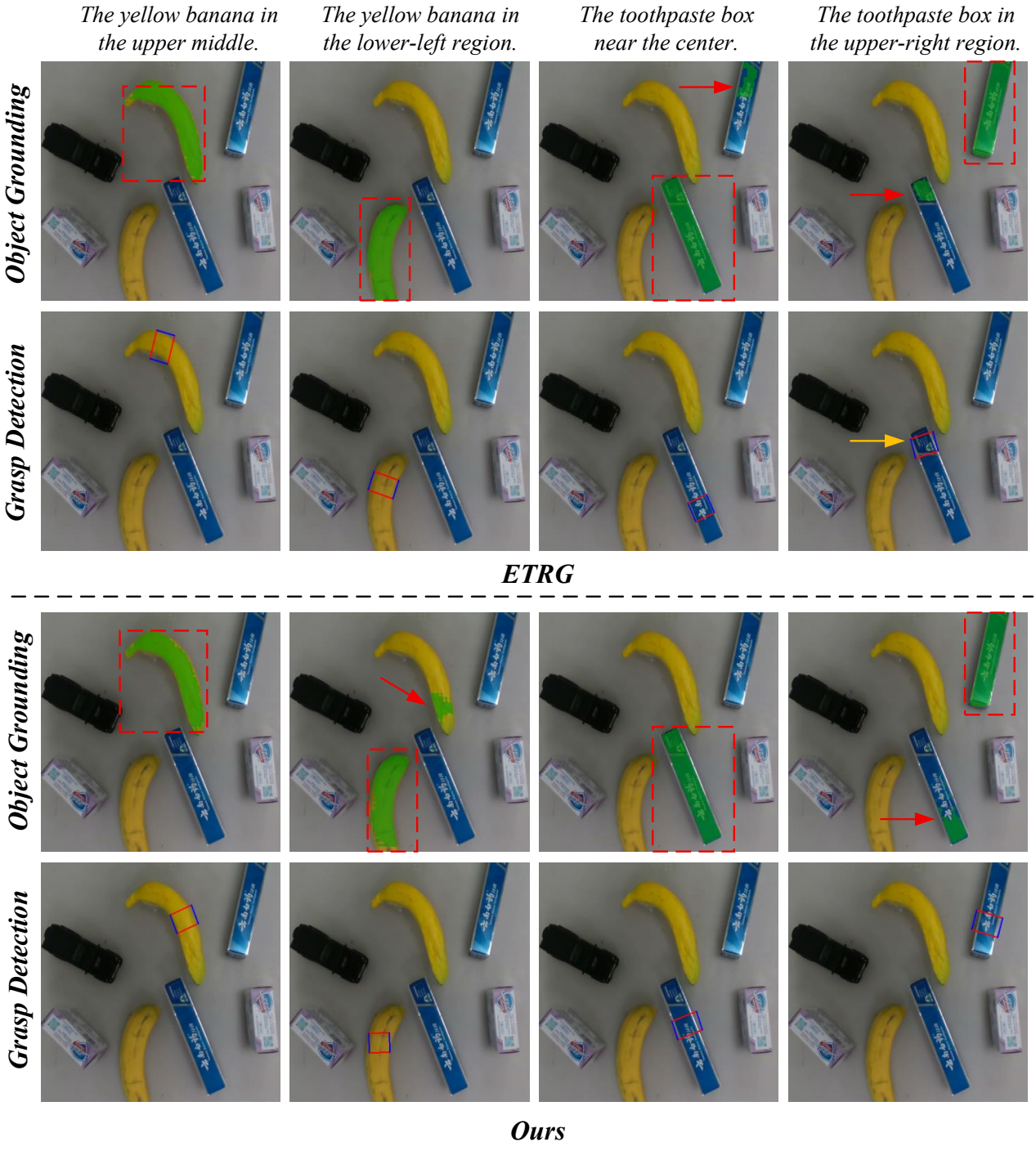


Figure 4: Qualitative comparison of object grounding and grasp detection results on Scene1 under different spatial expressions. The first two rows show the results of ETRG, while the last two rows show the results of our method. The referred target objects are highlighted by red dashed boxes, and the incorrect partial segmentation responses are indicated by red arrows.